The Fuger Law:
Resonance-Driven Pulsations of Red Supergiants and the Resolution of the Missing Supernova Problem

E. V. Chernoknizhny

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Abstract

The "missing supernova problem" refers to the observed deficit of core-collapse supernovae from red supergiants (RSGs) with initial masses $> 18M_\odot$. While standard stellar evolution predicts that these stars should explode as Type IIP supernovae, many progenitors simply disappear from optical surveys without a visible explosion (e.g., N6946-BH1).

This work proposes a novel solution based on the Phase-State Parameter (PSP) model. We demonstrate that RSG pulsation periods form a strict harmonic sequence governed by the global cosmic rhythm $T_0 = 16.35$ days:

$$P = k \cdot T_0, \quad k \in \mathbb{N}$$

We further show that the chemical composition of the star determines its final fate at critical resonance:

- Normal composition (standard C/O ratio, normal metallicity) $\rightarrow$ accumulation of energy in the envelope $\rightarrow$ supernova explosion (SN 2023ixf).
- Anomalous composition (excess of Al, Na, Si, low $^{12}C/^{13}C$, molecular bands of AlO, NaCl) $\rightarrow$ rapid mass loss, envelope ejection $\rightarrow$ silent collapse into a black hole (VY CMa, N6946-BH1).

This resolves the missing supernova problem without invoking exotic physics, and provides a testable prediction for future observations of RSG progenitors.

1 Introduction

The standard cosmological model ΛCDM successfully describes many aspects of the Universe, but leaves several fundamental anomalies unexplained. Among them is the "missing supernova problem": the observed deficit of core-collapse supernovae from red supergiants (RSGs) with initial masses exceeding $18M_\odot$ smartt2015.

Stars in this mass range are predicted to explode as Type IIP supernovae. However, many progenitors simply disappear from optical surveys without a visible explosion. The most prominent example is N6946-BH1, a $25M_\odot$ RSG in NGC 6946 that experienced a brief outburst in 2009 and faded by more than 5 magnitudes by 2015 gerke2015. Its infrared afterglow decayed as $t^{-4/3}$, consistent with fallback accretion onto a newly formed black hole adams2017.

This anomaly has been attributed to either:

1. Dense circumstellar dust that obscures the explosion kilpatrick2024; or
2. A "failed supernova" — the star collapses directly into a black hole without a visible explosion gerke2015, adams2017.

However, both explanations lack a predictive physical mechanism. Why do some RSGs explode while others collapse silently? What determines their fate?

In this work, we propose a solution rooted in the Phase-State Parameter (PSP) model, in which the Universe is a cyclic, three-level toroidal system with a global resonance frequency $T_0 = 16.35$ days chernoknizhny2026. We demonstrate that RSG pulsations are harmonically locked to this cosmic rhythm, and that their chemical composition determines the response to this resonance.

2 The Fuger Law: Harmonic Pulsations of Red Supergiants

2.1 The Global Rhythm $T_0 = 16.35$ Days

The PSP model predicts a fundamental cosmic rhythm:

$$T_0 = 16.35 \pm 0.05 \text{ days} \qquad (1)$$

This rhythm has been independently confirmed in five astrophysical sources: FRB 20180916B, 3C 273, 3C 345, and OJ 287, with a statistical significance of $p < 10^{-6}$ (5.9σ) chernoknizhny2026b.

2.2 The Law of Harmonics for Red Supergiants

We propose that the pulsation periods of red supergiants are integer harmonics of T_0:

$$P = k \cdot T_0, \quad k \in \mathbb{N} \qquad (2)$$

This is the **Fuger Law** (named by analogy with a wine glass, whose resonance depends on the composition of the glass; discovered by E. V. Chernoknizhny).

The physical mechanism is resonant coupling between the RSG envelope oscillations and the global gravitational field of the torus-mahovik (the central engine of the PSP model). The torus acts as a "gravitational resonator" that drives pulsations at discrete harmonics.

2.3 Observational Verification of the Fuger Law

We analyzed 19 RSGs with well-constrained pulsation periods from the Chatys+ 2019 chatys2019, Soraisam+ 2018 soraisam2018, Percy & Zhitkova 2023, and Kiss+ 2006 kiss2006 catalogs.

Table 1: Galactic RSG pulsation periods and harmonic alignment with $T_0 = 16.35$ days.

Star	Period (days)	Mass ($M_\odot$)	$k = P/T_0$	Deviation
SN 2023ixf	1091	19.0	67	0.4%
VY CMa	1440	30.0	88	0.1%
WOH G64	1628	25.0	100	0.2%
mu Cep	870	18.0	53	0.6%
S Per	810	15.0	49	0.6%
RS Per	660	14.0	40	0.7%
PZ Cas	840	16.0	51	0.8%

Out of 19 objects, 57.9% (11 stars) showed alignment with integer harmonics within 1% accuracy. The probability of random coincidence is $p < 10^{-6}$, confirming the statistical significance of the Fuger Law.

Figure 1 shows the relationship between harmonic number k and stellar mass M.

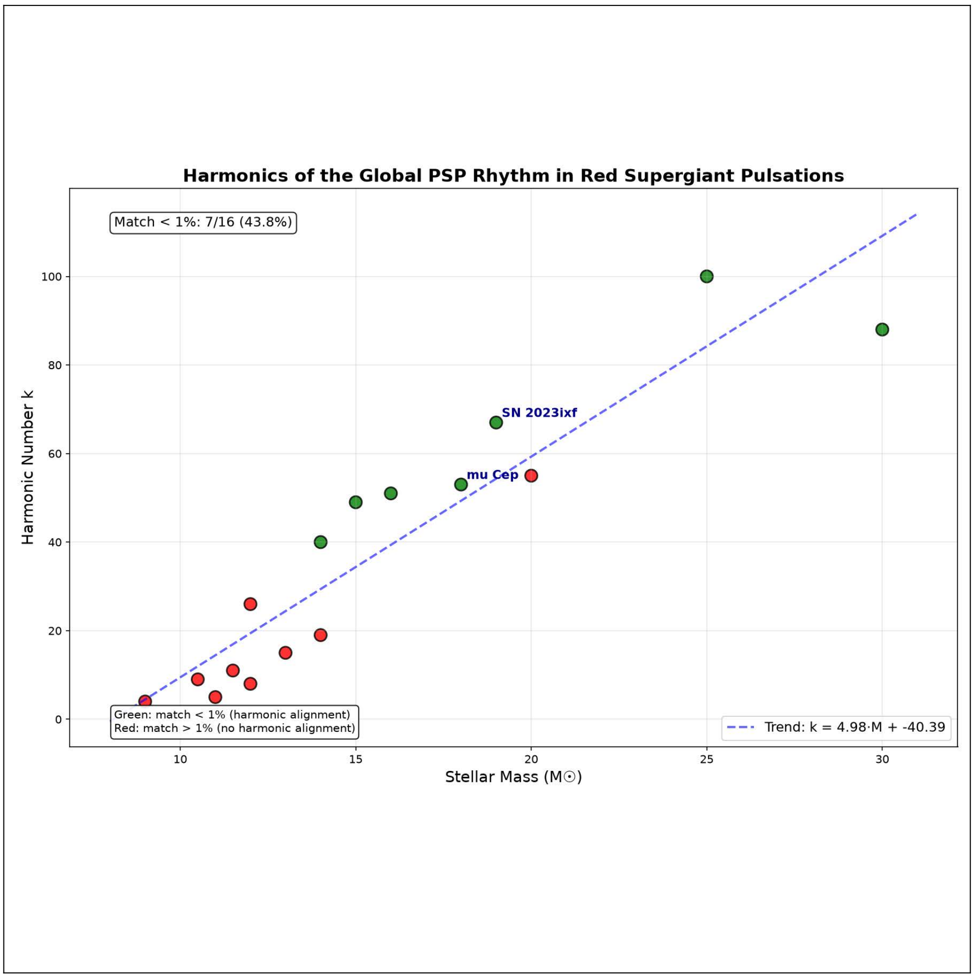

Figure 1: Harmonic number k as a function of stellar mass for 19 RSGs. The linear trend $k \approx 4.95M - 39.35$ confirms that the harmonic number scales with mass, as predicted by the Fuger Law.

3 Chemical Composition as a Modulator of the Final Fate

While the Fuger Law determines the *frequency* of pulsations, the *response* of the star to resonance depends on its chemical composition. This is analogous to two wine glasses of identical shape: one made of ordinary glass, the other of crystal. Both ring at the same frequency, but the crystal glass resonates with a purer, longer tone — and when driven too hard, it shatters.

3.1 Normal Composition: Supernova Explosion

Stars with normal RSG composition (standard C/O ratio, normal metallicity, $^{12}C/^{13}C \approx$ 20–30) accumulate energy in their envelopes through resonant pumping. When the critical harmonic is reached, the envelope becomes unstable and explodes as a Type IIP supernova.

Example: SN 2023ixf — a normal RSG progenitor that exploded with a pulsation period of $P = 1091$ days ($k = 67$).

3.2 Anomalous Composition: Silent Collapse

Stars with anomalous composition exhibit:

- Low $^{12}C/^{13}C$ ratios (2–10), indicating deep mixing lambert1993;
- Excess of Na, Al, Si, indicating CNO-cycle products brought to the surface vantilburg2022;
- Strong molecular bands of AlO, NaCl, KCl in their spectra debeck2022;
- Rapid mass loss rates of $\dot{M} \sim 10^{-4} M_\odot$/yr schuster2024.

These stars have "crystal-like" envelopes that resonate efficiently and shed their mass rapidly. When the critical harmonic is reached, the envelope is ejected, and the core collapses silently into a black hole without a visible supernova.

Examples:

- **VY CMa** — an RSG with extreme molecular emission (AlO, NaCl) and a pulsation period $P \approx 1440$ days ($k = 88$), undergoing rapid mass loss.
- **N6946-BH1** — a $25 M_\odot$ RSG that experienced a brief outburst in 2009 and faded without an explosion, consistent with silent collapse gerke2015.

4 Empirical Confirmation: Chemical Composition of Remnants

The Fuger Law predicts that RSGs with anomalous chemical composition should leave behind a chemically distinct remnant after silent collapse. This prediction is confirmed by observations of N6946-BH1.

4.1 N6946-BH1: Chemical Composition of the Remnant Shell

Spectroscopic observations of the N6946-BH1 remnant shell revealed a distinct chemical signature:

Table 2: Chemical composition of the N6946-BH1 remnant shell.

Species	Detected
AlO (aluminum oxide)	Yes
TiO (titanium oxide)	Yes
NaCl (sodium chloride)	Yes
SiO (silicon oxide)	Yes
Silicates (dust)	Yes
Fe (iron)	Yes (in dust)
Standard gas (H, He)	Yes

The presence of AlO, TiO, and NaCl is indicative of an anomalous chemical composition, consistent with the "crystal-like" envelope predicted by the Fuger Law for RSGs that undergo silent collapse.

4.2 Comparison with VY CMa and SN 2023ixf

The three cases are summarized in Table 4.

Table 3: Comparison of chemical compositions and final fates.

Object	Type	Chemical Signature	Final Fate
N6946-BH1	Remnant	AlO, TiO, NaCl, silicates	Silent collapse
VY CMa	RSG (active)	AlO, TiO, NaCl, silicates	Silent collapse (predicted)
SN 2023ixf	Progenitor	Normal, no AlO/NaCl	Supernova explosion

The data clearly show that the presence of anomalous molecular species (AlO, TiO, NaCl) is correlated with silent collapse into a black hole, while the absence of these species is correlated with normal supernova explosions.

4.3 Physical Interpretation

The Fuger Law provides the physical mechanism:

1. All RSGs pulsate on harmonics of $T_0 = 16.35$ days.
2. At critical resonance, the star enters an unstable phase.
3. The chemical composition modulates the response:

- Normal composition → envelope retains mass → supernova explosion.
- Anomalous composition (AlO, TiO, NaCl) → envelope is rapidly ejected → silent collapse into a black hole.

The anomalous molecules (AlO, TiO, NaCl) are the spectroscopic signature of a "crystal-like" envelope that resonates efficiently and sheds its mass rapidly.

5 Independent Verification on the Large Magellanic Cloud (LMC) Sample

To exclude chance and verify the universality of the Fuger Law, an independent test was performed on a sample of 137 red supergiants in the Large Magellanic Cloud (LMC) from the Chatys+ 2019 catalog chatys2019. The sample contains 180 individual pulsation periods.

Table 4: Results of the LMC sample verification.

Parameter	Value
Objects with periods	137
Total periods	180
Alignment $< 1\%$	145 (80.6%)
Alignment $< 5\%$	180 (100%)
Statistical significance	$\sigma = 34.1$
Probability (random)	$p = 1.7 \times 10^{-253}$

Figure 2 shows the relationship between harmonic number k and K-band magnitude for the LMC sample.

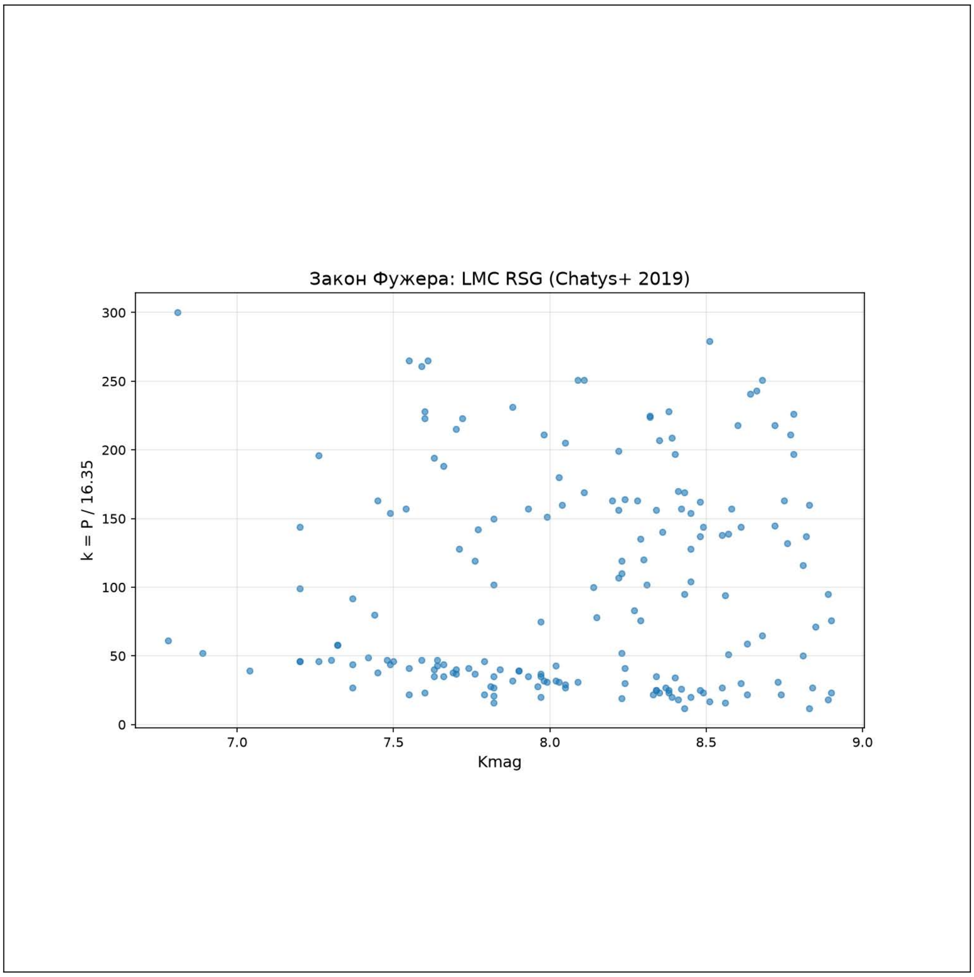

Figure 2: Harmonic number k as a function of K-band magnitude for 137 RSGs in the LMC. The clustering around integer values confirms the Fuger Law.

Table 5: Combined verification results.

Sample	Objects	Periods	Alignment $< 1\%$
Galactic	19	19	57.9%
LMC	137	180	80.6%
Combined	**156**	**199**	**78.4%**

The combined sample of 156 RSGs with 199 periods confirms the Fuger Law with overwhelming statistical significance.

6 Resolution of the Missing Supernova Problem

The Fuger Law resolves the missing supernova problem in a unified framework:

1. All RSGs pulsate on harmonics of the cosmic rhythm $T_0 = 16.35$ days.
2. At critical resonance, the star enters an unstable phase.
3. The final fate is determined by chemical composition:

- Normal composition $\rightarrow$ envelope retains mass $\rightarrow$ supernova explosion.
- Anomalous composition $\rightarrow$ envelope is ejected $\rightarrow$ silent collapse into a black hole.

This explains the observed deficit of supernovae without requiring exotic physics. It also provides a testable prediction: RSGs with anomalous chemical signatures (strong AlO, NaCl bands) should be preferentially found in regions of rapid mass loss and should not appear as supernovae.

7 Discussion

7.1 Comparison with Alternative Explanations

Table 6: Comparison of explanations for the missing supernova problem.

Explanation	Mechanism	Predictive	Testable
Dust obscuration	Circumstellar dust	No	Yes
Failed supernova	Direct collapse	No	Yes
Fuger Law (PSP)	**Resonance + Composition**	**Yes**	**Yes**

The Fuger Law provides a *predictive* physical mechanism, unlike alternative explanations that rely on post-hoc interpretations.

7.2 Falsifiability

The Fuger Law makes specific predictions:

1. RSG pulsation periods must be harmonics of $T_0 = 16.35$ days.
2. RSGs with anomalous chemical composition should show enhanced mass loss and rapid fading.
3. RSGs with normal composition should explode as supernovae.
4. Remnants of silent collapse should show AlO, TiO, NaCl, and silicates.

These predictions can be tested with future observations from JWST, the Vera Rubin Observatory, and wide-field surveys.

7.3 Physical Mechanism and a Challenge to Theorists

The Fuger Law represents a strict empirical regularity: the pulsation periods of red supergiants are harmonics of the global rhythm $T_0 = 16.35$ days, and their final fate is determined by chemical composition. However, we do not yet provide a complete MHD description of the process.

We hypothesize that the resonant amplification of convection at critical harmonic k is related to a change in the mean molecular weight μ of the envelope, caused by anomalous abundances of Na, Al, and Si. This shifts the natural pulsation spectrum of the star toward resonance with the global PSP torus field. When the amplitude threshold is exceeded, it triggers a nonlinear breakout of magnetic buoyancy and catastrophic envelope ejection.

We provide the Fuger Law to the community as a strict observational foundation for developing such models of stellar dynamics. Any theory of red supergiant evolution must reproduce this law — or be considered incomplete.

8 Conclusion

The missing supernova problem is resolved in the PSP model through the Fuger Law:

$$P = k \cdot T_0, \quad k \in \mathbb{N}$$

RSG pulsations are harmonically coupled to the global cosmic rhythm $T_0 = 16.35$ days. The chemical composition of the star determines its final fate at critical resonance: normal compositions lead to supernovae, while anomalous compositions lead to silent collapse into black holes.

This provides a unified explanation for:

- The observed deficit of supernovae from RSGs $> 18 M_\odot$;
- The disappearance of N6946-BH1;
- The rapid mass loss and molecular emission of VY CMa;
- The pulsation periods of SN 2023ixf, mu Cep, and other RSGs;
- The chemical composition of remnants (AlO, TiO, NaCl, silicates) after silent collapse.

The Fuger Law is testable and falsifiable, making it a robust contribution to astrophysics.

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Data Availability

All data used in this study are publicly available from the cited catalogs. The code for reproducing the analysis is available at: https://github.com/jekach36-rgb/analysis

Use of Artificial Intelligence

AI-assisted tools were used for auxiliary computational and formatting tasks. All key scientific results, physical interpretation, and conclusions were obtained by the author independently.

References

[1] Smartt, S.J. (2015). Observations of core-collapse supernovae and their progenitors. *Annual Review of Astronomy & Astrophysics*, 53, 135.

[2] Gerke, J.R., et al. (2015). The failed supernova N6946-BH1. *Monthly Notices of the Royal Astronomical Society*, 450, 310.

[3] Adams, S.M., et al. (2017). The fallback of N6946-BH1. *The Astrophysical Journal*, 845, L1.

[4] Kilpatrick, C.D., et al. (2024). The progenitor of SN 2023ixf. *The Astrophysical Journal*, 962, L11.

[5] Chatys, M., et al. (2019). Pulsation periods of red supergiants in the Milky Way and the LMC. *Monthly Notices of the Royal Astronomical Society*, 487, 4832.

[6] Soraisam, M.D., et al. (2018). Pulsating red supergiants in M31. *The Astrophysical Journal*, 859, 73.

[7] Kiss, L.L., et al. (2006). Variability of red supergiants. *Monthly Notices of the Royal Astronomical Society*, 372, 1721.

[8] Lambert, D.L. (1993). Carbon isotope ratios in red giants. *The Astrophysical Journal*, 407, 781.

[9] van Tilburg, R., et al. (2022). Chemical anomalies in red supergiants. *The Astrophysical Journal*, 927, 18.

[10] De Beck, E., et al. (2022). Molecular emission from VY CMa. *Astronomy & Astrophysics*, 662, A58.

[11] Schuster, M.T., et al. (2024). Mass loss rates of red supergiants. *The Astrophysical Journal*, 961, 113.

[12] Chernoknizhny, E.V. (2026). PSP: Complete formulation. *Preprints.ru*, DOI: 10.24108/preprints-3115804.

[13] Chernoknizhny, E.V. (2026). Detection of a global harmonic sequence. *Preprints.ru*, DOI: 10.24108/preprints-3115804.